6.4.3 System Architectural Design Process

The **System Architectural Design Process** in this International Standard is a specialization of the **Architectural Design Process** of ISO/IEC 15288. Users may consider claiming conformance to the ISO/IEC 15288 process instead of this process.

6.4.3.1 Purpose

The purpose of the System Architectural Design Process is to transform the system requirements into a structured system architecture. This architecture identifies system elements (hardware, software, and manual operations), allocates requirements to these elements, and defines their internal and external interfaces. It provides a coherent framework that supports implementation, integration, verification, and validation throughout the life cycle.

6.4.3.2 Outcomes

As a result of successful implementation of the System Architectural Design Process:

a) A system architecture design is developed that identifies system elements and meets the defined system requirements.
b) Functional and non-functional requirements (including performance, security, safety, and environmental considerations) are addressed in the system architecture.
c) System requirements are allocated to system elements in a structured and traceable manner.
d) Internal and external interfaces of each system element are defined.
e) Verification between system requirements and the system architecture is performed.
f) Bidirectional traceability between allocated requirements and the customer’s requirements baseline is established and maintained.
g) Consistency and traceability between system requirements and the system architecture design are maintained.
h) The system requirements, architecture design, and their relationships are baselined and communicated to all affected parties.
i) Human factors and ergonomic principles are incorporated in the architecture, including identification and execution of human-centred design activities.

6.4.3.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.3.3.1 Establishing the System Architecture

This activity develops a top-level architecture and allocates requirements to system elements.

- 6.4.3.3.1.1 Develop a top-level system architecture that identifies the system elements, including hardware, software, and manual operations.
- 6.4.3.3.1.2 Allocate all system requirements among the identified system elements, ensuring completeness and avoiding ambiguity or overlap.
- 6.4.3.3.1.3 Define internal and external interfaces of each system element in the architecture, including data, control, and human interfaces.
- 6.4.3.3.1.4 Incorporate human factors and ergonomic knowledge into system design and identify required human-centred design activities.
- 6.4.3.3.1.5 Document the system architecture and allocated system requirements.
- 6.4.3.3.1.6 Baseline and communicate the system architecture and its relationship with system requirements to all affected parties.

Cross-reference: This activity builds upon the **System Requirements Analysis Process (6.4.2)** and provides the basis for **System Implementation (6.4.4)** and **Software Architectural Design (7.1.3)**.

6.4.3.3.2 Architectural Evaluation

This activity evaluates the system architecture to ensure its adequacy, consistency, and feasibility.

- 6.4.3.3.2.1 Evaluate the system architecture and the allocated requirements using defined criteria, including:
- Bidirectional traceability to system and stakeholder requirements.
- Consistency with system requirements.
- Appropriateness of design standards, architectural principles, and methods applied.
- Feasibility of software, hardware, and manual elements in meeting their

allocated requirements.

- Compliance with performance, safety, security, usability, environmental, and other non-functional requirements.
- Feasibility of system operation, integration, verification, and maintenance.
- 6.4.3.3.2.2 Document evaluation results and address identified deficiencies through architectural refinement as required.
- 6.4.3.3.2.3 Maintain evaluation records for traceability and audit purposes.

Cross-References

- **System Requirements Analysis Process (6.4.2)** — source of system requirements for architectural design.
- **System Implementation Process (6.4.4)** — uses the architecture as the basis for realization.
- **Software Architectural Design Process (7.1.3)** — derives the software architecture from system architecture.
- **Verification (7.2.4)** — checks conformance between architecture and system requirements.